LF5 Worked Examples: Linear Models

Recording companion for the LF5 CheckIt support video — Math 172, Santa Ana College

How to use this document

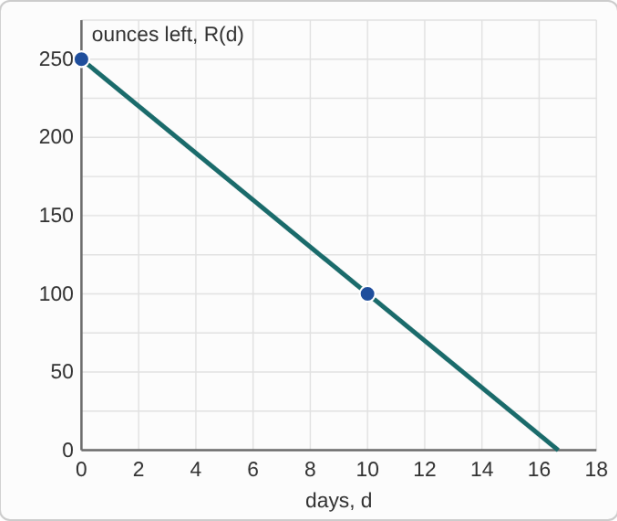
- Each worked example is on its own page, in the same order as the Try It tabs in the interactive explainer, with the same numbers.
- The right-hand third of every page is intentionally blank. It is space for handwriting or ink notation while you narrate (for example, after inserting this PDF as a printout in OneNote).
- Say it at the start of the video: your CheckIt problem will have different numbers than the examples here. The reasoning is what transfers.

The one question behind every example

What is the starting amount, and what changes by the same amount each time? A linear model is a starting amount plus a constant rate times the input: $f(x) = mx + b$. The rate is the slope, with a minus sign if the amount goes down. The starting amount is the y-intercept, the output when the input is 0. Before you plug in or solve, check what x counts: items, additional lines, days, or years after the start year.

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Example 1: A Bag of Rice



Plotted data: A graph with days, d, from 0 to 18 across and ounces left, R(d), from 0 to 250 up. A straight line starts at (0, 250), marked with a dot, falls through the marked point (10, 100), and reaches 0 between day 16 and day 17.

Question

A family buys a 250-ounce bag of rice and uses 15 ounces per day. R(d) is the rice left after d days. What is the slope of R, and what does it mean?

- **A.** 250 ounces: the amount in the bag at the start
- **B.** −15 ounces per day: the rice left drops 15 ounces a day
- **C.** 250 ounces per day: the family uses 250 ounces each day
- **D.** 15 days: how long the bag of rice lasts before it is empty

Answer and reasoning

The model is $R(d)=250-15d$. Each day the rice left goes down by 15 ounces, so the slope is -15, in ounces per day. On the graph, the line drops 150 ounces over 10 days: $(-150)/10=-15$. CheckIt's answer key prints this slope as 15, the rate of use; say both: the family uses 15 ounces a day, so the slope of R is -15. The 250 ounces is the y-intercept, not the slope; 250 ounces per day treats the starting amount as the rate; and

the bag lasts $250/15 \approx 16.7$ days, not 15, because 15 is the ounces used per day. **-15 ounces per day.**

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Example 2: A Phone Plan

$$C(x)=10x+35$$

Question

A phone plan costs $35 a month for the first line and $10 for each additional line. C(x) is the monthly cost with x additional lines. What is the y-intercept, and what does it mean?

A. 35: the cost with only the first line
B. 10: the starting cost, before any lines are added
C. 45: the cost of the first line, since x = 1
D. 0: a plan with no lines costs nothing

Answer and reasoning

The model is $C(x)=10x+35$, and the y-intercept is $C(0)=35$. Here x counts additional lines, so $x=0$ is a plan with just the first line, which costs $35. The $10 for each additional line is the slope, the rate, not the starting cost; $C(1)=45$ is the plan with one additional line, two lines in all, because $x=1$ does not mean the first line; and $x=0$ still includes the first line, so the cost is not 0. **35: the cost with only the first line.**

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Example 3: A Shrinking Town

$$P(x) = -1500x + 48000$$

Question

In 2010, the population of Pawnee was 48,000, and it has decreased by 1,500 per year since. P(x) is the population x years after 2010. What does the y-intercept of P mean?

A. The population in the calendar year 0
B. The number of people the town loses each year
C. The population in 2011, a year after the start
D. The population in 2010, when the model starts

Answer and reasoning

The y-intercept is $P(0) = 48{,}000$. Since x counts years after 2010, $x = 0$ is the year 2010 itself, so 48,000 is the population in 2010. The calendar year 0 reads x as the year instead of the years since 2010; the people lost each year, 1,500, is the slope, $-1{,}500$ people per year; and the population in 2011 is $P(1) = 46{,}500$, which treats the start year as $x = 1$. **The population in 2010, 48,000 people.**

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Example 4: Reading a Table

Table 1. Population of Hill Valley x years after 2005

Years after 2005, x	0	1	2	3
Population, P(x)	64,000	61,500	59,000	56,500

Question

The population of Hill Valley changes by the same amount each year. What is the slope of the linear model P, and what does it mean?

A. −2,500 people per year: the town loses 2,500 a year
B. 64,000 people: the population of the town in 2005
C. 61,500 people: the population after one year
D. −7,500 people per year: the change over the years shown

Answer and reasoning

Each step of 1 year changes the population by $61{,}500-64{,}000=-2{,}500$, so the slope is $-2{,}500$ people per year, and $P(x)=-2{,}500x+64{,}000$. For a shrinking town, CheckIt's answer key prints the slope as negative too. 64,000 is $P(0)$, the y-intercept; 61,500 is $P(1)$, an amount, while the slope is a change per year; and $-7{,}500$ is $56{,}500-64{,}000$, the change over 3 years, not yet divided by the 3. **−2,500 people per year.**

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Example 5: Widgets

$$C(x)=mx+b$$

Question

Manufacturing widgets costs $12 per widget plus an additional $400 for fixed costs. Which function gives C(x), the cost of manufacturing x widgets?

A. $C(x)=400x+12$
B. $C(x)=12x+400$
C. $C(x)=12x-400$
D. $C(x)=412x$

Answer and reasoning

The rate is $12 per widget, so the slope is 12; the fixed costs, $400, are paid even for 0 widgets, so they are the y-intercept. $C(x)=12x+400$. Check: 0 widgets cost $C(0)=400$, the fixed costs. $400x+12$ swaps the rate and the starting amount; $12x-400$ subtracts the fixed costs, but they add to the cost; and $412x$ adds the two numbers, which charges the fixed costs again for every widget. ***C(x)=12x+400*.**

Example 6: Population

$$P(x)=mx+b$$

Question

In 2015, the population of Springfield was 85,000, and it has decreased by 3,000 per year since. Which function gives P(x), the population x years after 2015?

A. $P(x)=85000x+3000$
B. $P(x)=85000-3000x$
C. $P(x)=85000x-3000$
D. $P(x)=85000-3000(x-2015)$

Answer and reasoning

The town starts at 85,000 when $x=0$, the year 2015, and loses 3,000 people each year, so the slope is $-3{,}000$: $P(x)=85{,}000-3{,}000x$. $85000x+3000$ makes two slips, swapping the numbers and making the town grow; $85000x-3000$ swaps the rate and the starting amount; and $85000-3000(x-2015)$ treats x as the calendar year, but here x is years after 2015, so it would give $P(0)=6{,}130{,}000$. ***P(x)=85,000−3,000x*.**

Example 7: Phone Lines

$$C(x)=mx+b$$

Question

A phone plan costs $25 a month for the first line and $15 for each additional line. Which function gives C(x), the monthly cost with x additional lines?

- **A.** $C(x)=15x+25$
- **B.** $C(x)=25x+15$
- **C.** $C(x)=15(x-1)+25$
- **D.** $C(x)=40x$

Answer and reasoning

The first line is always there, so $25 is the starting amount, and each additional line adds $15: $C(x)=15x+25$. Check: one additional line, two lines in all, costs $C(1)=40$. $25x+15$ swaps the two prices; $15(x-1)+25$ would be right if x counted all the lines, but x counts only the additional ones; and $40x$ charges both prices for every additional line. ***C(x)=15x+25*.**

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Example 8: A Leaking Tank

$$W(x)=mx+b$$

Question

A leaky tank is losing water at a constant rate of 4 ounces per hour. The tank started with 200 ounces. Which function gives W(x), the water left after x hours?

A. $W(x)=4x+200$
B. $W(x)=200x-4$
C. $W(x)=200-4x$
D. $W(x)=4x-200$

Answer and reasoning

The tank starts with 200 ounces and loses 4 each hour, so the slope is -4: $W(x)=200-4x$. Check: after 10 hours, $200-40=160$ ounces are left. $4x+200$ adds water instead of losing it; $200x-4$ swaps the rate and the starting amount; and $4x-200$ subtracts in the wrong order, which starts the tank at -200 ounces. ***W(x)=200−4x*.**

Example 9: Population Later

$$P(x)=2000x+34000$$

Question

In 2012, the population of Capeside was 34,000, and it has grown by 2,000 per year since. P(x) is the population x years after 2012. What will the population be in 2027?

A. 4,088,000
B. 64,000
C. 30,000
D. 62,000

Answer and reasoning

The year 2027 is $x=2027-2012=15$ years after the start, so $P(15)=2{,}000(15)+34{,}000=64{,}000$. Putting in $x=2027$ gives 4,088,000, because x is years after 2012, not the year; 30,000 is only the growth, $2{,}000(15)$, without the starting population; and 62,000 uses 14 years, counting from 2013 instead of 2012. **64,000.**

Example 10: How Many Hats

$$C(x)=9x+540$$

Question

Producing hats costs $9 per hat plus an additional $540 for fixed costs. How many hats can be produced for $1,890?

A. 150 hats
B. 210 hats
C. 270 hats
D. 1,350 hats

Answer and reasoning

This time the output is known: solve $9x+540=1890$. Subtract 540: $9x=1350$. Divide by 9: $x=150$. Check: $9(150)+540=1890$. 210 is $1890/9$, which forgets the fixed costs; 270 is $(1890+540)/9$, which adds 540 instead of subtracting it; and 1,350 stops before dividing by 9, so it is the dollars left for hats, not the number of hats. **150 hats.**

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Example 11: Water Left

$$W(x)=-5x+55$$

Question

A leaky bucket started with 55 ounces of water and loses 5 ounces per minute. After how many minutes will the bucket contain 15 ounces of water?

A. 3 minutes
B. 14 minutes
C. 8 minutes
D. 11 minutes

Answer and reasoning

Solve $-5x+55=15$: subtract 55, $-5x=-40$, and divide by -5, $x=8$. Check: $55-5(8)=15$. 3 minutes is $15/5$, which divides the water left instead of the water lost; 14 minutes is $(55+15)/5$, which adds the 15; and 11 minutes is $55/5$, when the bucket is empty, which answers a different question. **8 minutes.**

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Example 12: Lines for a Budget

$$C(x)=10x+20$$

Question

A phone plan costs $20 a month for the first line and $10 for each additional line. How many additional lines can be purchased for $50?

A. 5
B. 3
C. 4
D. 2

Answer and reasoning

Solve $10x+20=50$: $10x=30$, so $x=3$ additional lines, 4 lines in all. Check: $10(3)+20=50$. 5 is $50/10$, which forgets the first line's $20; 4 is the number of lines in all, but the question asks for additional lines, which is what x counts; and 2 subtracts the first line again, as if $x=3$ counted it. **3 additional lines.**

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Example 13: Slip 1

$$W(x)=3x+55$$

Question

A leaky bucket starts with 55 ounces of water and loses 3 ounces per minute. A student wrote this model for W(x), the water left after x minutes. What is the mistake, if any?

A. There is no mistake.
B. Water is lost, so the slope is −3: W(x) = 55 − 3x.
C. The rate and the start are swapped: it should be W(x) = 55x + 3.
D. Subtract the 55 instead: W(x) = 3x − 55, since water is lost.

Answer and reasoning

The bucket loses water, so the amount goes down 3 ounces each minute and the slope is -3: $W(x)=55-3x$. The student's model has the bucket gaining water. $55x+3$ swaps the rate and the starting amount; and $3x-55$ subtracts in the wrong order, which starts the bucket at -55 ounces. ***$W(x)=55-3x$*.**

Example 14: Slip 2

$$4000x+72000=92000$$
$$4000x=20000$$
$$x=5$$

Question

In 2018, the population of Bayside was 72,000, and it has grown by 4,000 per year since. P(x) is the population x years after 2018. A student found the year when the population will be 92,000 and answered "5". What is the mistake, if any?

A. There is no mistake.

B. x is years after 2018, so the year is 2023.

C. Divide 92,000 by 4,000: x = 23, so the year is 2041.

D. 2018 is the first year, so the year is 2022.

Answer and reasoning

The solving is right: $x=5$. But x is years after 2018, and the question asks for a year: $2018+5=2023$. Check: $P(5)=20{,}000+72{,}000=92{,}000$. CheckIt's answer key gives the calendar year. 23 is $(92{,}000)/(4{,}000)$, which forgets the starting population; and 2018 is $x=0$, not year 1, so 5 years later is 2023, not 2022. **The year 2023.**

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Example 15: Slip 3

$$C(x)=10x+40$$
$$C(3)=10(3)+40=70$$

Question

A phone plan costs \$40 a month for the first line and \$10 for each additional line. A student found the monthly cost with 3 additional lines. What is the mistake, if any?

A. There is no mistake.
B. The first line is not extra: use C(x) = 10(x − 1) + 40.
C. Three additional lines means x = 4, so the cost is \$80.
D. The slope should be 40, the bigger of the two prices.

Answer and reasoning

The first line costs \$40 whatever happens, so it is the starting amount, and each additional line adds \$10: $C(x)=10x+40$. With 3 additional lines, $C(3)=70$, which is 4 lines in all. The student is right. $10(x-1)+40$ would be right if x counted every line, but here x counts only the additional ones; for the same reason, 3 additional lines is $x=3$, not 4; and the slope is the price that repeats, \$10, not the larger price. **There is no mistake.**

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Example 16: Slip 4

$$C(x)=7x+600$$

Question

Manufacturing books costs $7 per book plus an additional $600 for fixed costs. A student wrote this model and said, "The y-intercept, 600, is the cost of manufacturing one book." What is the mistake, if any?

A. There is no mistake.
B. The y-intercept is 7, the cost of each book.
C. 600 is the cost of 0 books: the fixed costs.
D. 600 is the number of books that can be made, not a cost.

Answer and reasoning

The model is right, but the explanation is not. The *y*-intercept is the output when the input is 0: $C(0)=600$, the fixed costs paid before any books are made. One book costs $C(1)=607$. The 7 is the slope, the cost per book, not the *y*-intercept; and $C(x)$ is a cost in dollars, so 600 is dollars, not a number of books. **600 dollars is the cost of making 0 books: the fixed costs.**

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Example 17: Slip 5

$$180-9d=54$$
$$-9d=234$$
$$d=-26$$

Question

A family buys a 180-ounce bag of rice and uses 9 ounces per day. A student found when 54 ounces will be left. What is the mistake, if any?

A. There is no mistake.
B. Set the model equal to 0 instead, which gives d = 20 days.
C. Just divide the 54 by the 9 ounces a day: d = 6 days.
D. Subtracting 180 gives −9d = −126, so d = 14.

Answer and reasoning

The model $R(d)=180-9d$ and the equation are right. Then subtract 180 from both sides: $54-180=-126$, so $-9d=-126$ and $d=14$. The student added 180 instead, $54+180=234$. A negative number of days is a sign that something went wrong. Check: $180-9(14)=54$. Setting the model equal to 0 answers a different question, when the bag is empty; and $54/9=6$ divides the amount left instead of the amount used. **14 days.**